

Math Circle!

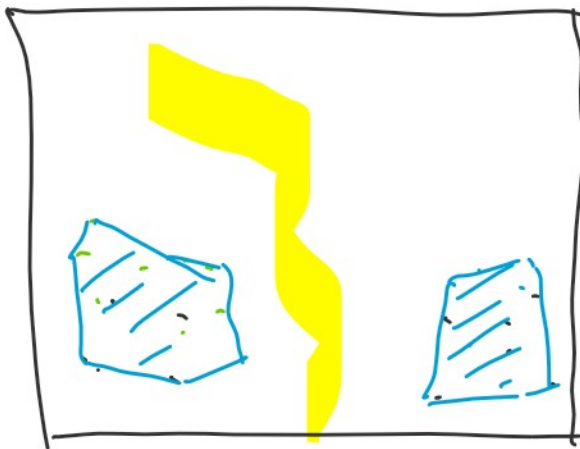
Saturday, May 14, 2022 1:26 PM

• The Idea of Divide and Conquer

- Robotics and convex hulls
 - Cryptography
-

Imagine a robot navigating obstacles...

What a robot sees...



Basic computational geometry idea:

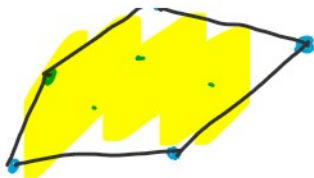
Convex Hull.

Puzzle: Given points

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

how do you find their convex hull?



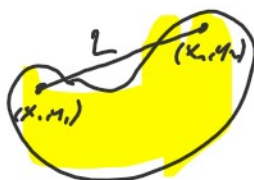


Def: A shape S in the plane $\mathbb{R}^2$ is convex if and only if for any points

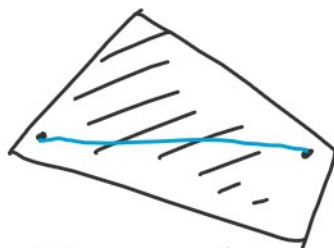
$(x_1, y_1), (x_2, y_2) \in S$ we have the

line segment L (connecting (x_1, y_1) & (x_2, y_2))

is also in S .



NOT convex!



IS convex!



(circle, NOT disk!)

NOT convex!



Disk is convex!

Some polygons are convex, not all!



Not convex!

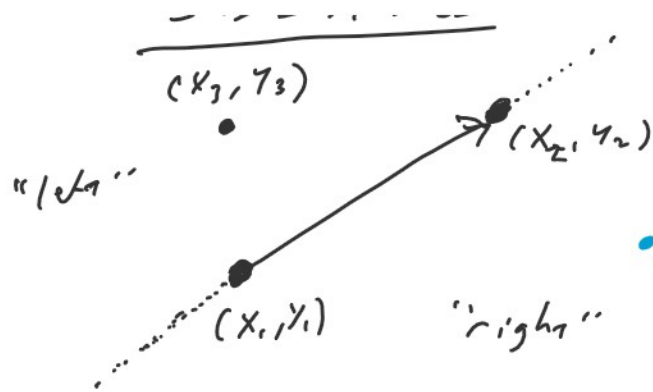


KEY: ... to programming this is to know SIDE DNESS!

(x_3, y_3)



which

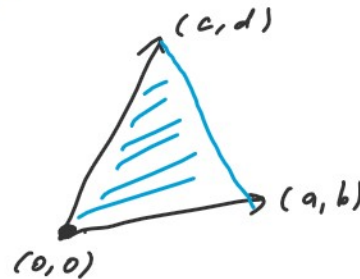


Which side are you on?

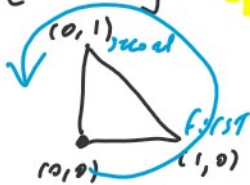
Trick: How can we measure area of

$\frac{1}{2}$. The determinant!

$$\frac{1}{2} \text{Det} \begin{bmatrix} a & b \\ c & d \end{bmatrix} := \frac{1}{2}(ad - bc).$$



Example:



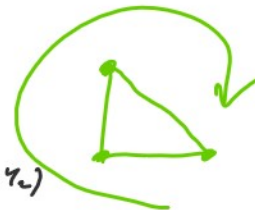
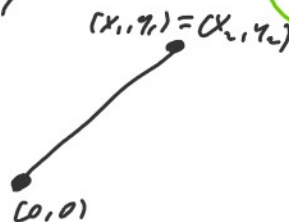
$$\text{area} = \frac{1}{2} = \frac{1}{2} \text{Det} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{2} \cdot (1 \cdot 1 - 0 \cdot 0)$$

What if you switch points?

$$\frac{1}{2} \text{Det} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \frac{1}{2}(0 - 1) = -\frac{1}{2}.$$

Signed area can be negative!

Now, what is the area of

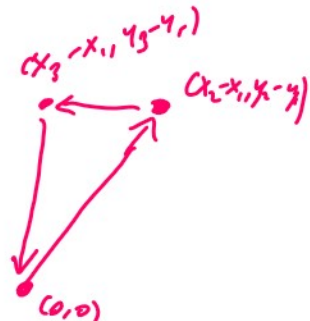
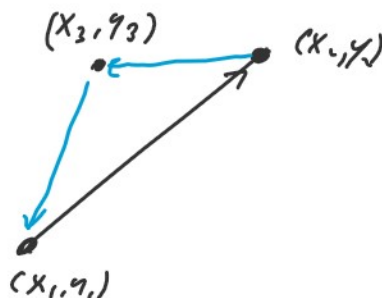


? $\rightarrow \underline{0}$

$$\text{Also, } \frac{1}{2} \text{Det} \begin{bmatrix} x_1 & y_1 \\ x_1 & y_1 \end{bmatrix} = \frac{1}{2}(x_1 y_1 - x_1 y_1) = 0.$$

Now we solve sidedness:

Consider the oriented triangle:



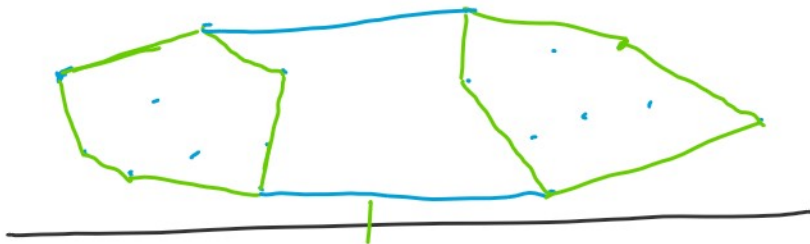
In other words, looking $\text{Sign} \left(\text{Det} \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{bmatrix} \right)$!

$+$ = left, $-$ = right

$+$ = left, $-$ = right

So how do we compute convex hulls?

Divide and Conquer is one way....



- ① Split points into left half & right half.
- ② Compute those convex hulls.
- ③ Merge.

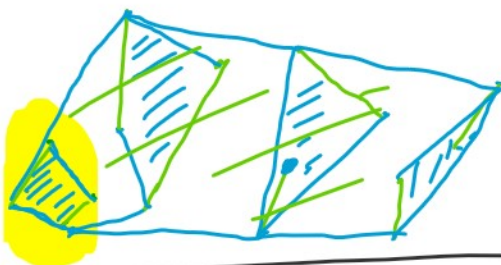
More concretely, you can first sort the points by x-coordinate, and then compute convex hulls of adjacent pairs!

Then merge the pairs...

Then merge the pairs of pairs...

etc...

Example:



pairs.

Activity: Write down

16 points in the plane
and perform this algorithm!

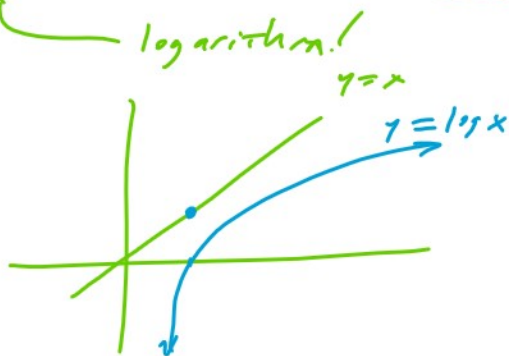
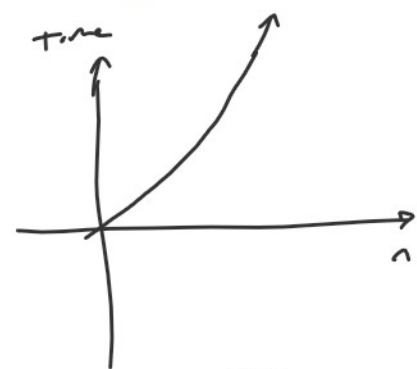
Convex hull (in the plane) is one of the few computational problems that is (almost) completely understood, from a complexity point of view...

if (almost) completely unnecessary...
the complexity point of view...

Fact #1: On any modern
commercial computer, convex hull
of n points in $\mathbb{R}^2$ will take time

$$\leq \text{Constant} \times n \log n$$

base-10
is OK...



$$O(n \log n) \quad (\text{upper bound})$$

What about lower bounds?

Can anyone do better? No!

Fact #2: On any modern computer you

can buy, NO algorithm will go
faster than ^(positive) Constant $\times n \log n$ (!)

(1970s...)

Fact #3: Convex hull and
Sorting have the "same"
complexity...

Sorting ...

Complexity ...

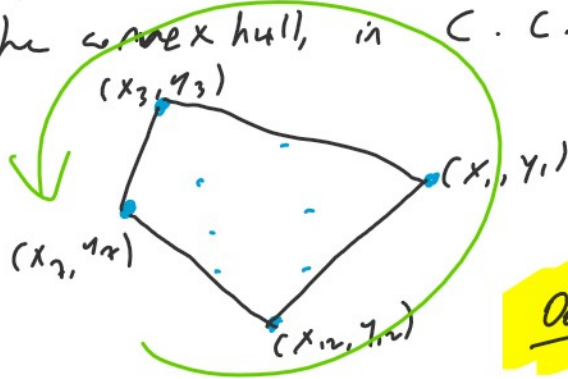
In fact, you can sort it you
can compute convex hulls!

Clarification: A proper convex hull
algorithm takes as input points

$$(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{Q}^2$$

(pairs of
rationals)

The output is a list of
indices for the points on the boundary
of the convex hull, in C.C.W. order ...



Out: 12, 1, 3, 7

How do you sort from convex hull?

Trick: Given $x_1, \dots, x_n$, to sort
via convex hull, do this!

Find convex hull of $(x_1, x_1^2), (x_2, x_2^2), \dots, (x_n, x_n^2)$.

